

Assignment#2: Advanced Glucose Estimation Model

Objective: Expand the baseline model (Age and BMI) by adding new physiological metrics to predict **Glucose** levels more accurately.

Task 1: The Expanded Mathematical Equation

In the baseline model, we used two features. For this assignment, you must include **Blood Pressure (BP)** and **Insulin Levels** as additional inputs.

Your Task: Write the new mathematical equation for this model.

The expanded equation should follow this structure:

$$\text{Glucose} = b_0 + b_1(\text{Age}) + b_2(\text{BMI}) + b_3(\text{Blood Pressure}) + b_4(\text{Insulin})$$

Task 2: Coding the Multi-Feature Model

Using the `glucose_multiple_linear_regression_explained.ipynb` as a template, modify **Step 1** and **Step 2** to include the new variables.

Code Update Task:

Update your feature selection in Python. Assuming your new dataset (`expanded_health_data.csv`) has the columns Age, BMI, Blood_Pressure, and Insulin:

Python

Python



```
# Step 1: Update input variables (X) to include 4 features
X = df[["Age", "BMI", "Blood_Pressure", "Insulin"]]
y = df["Glucose"]

# Step 2: Re-train the model
model = LinearRegression()
model.fit(X_train, y_train)

print("New Coefficients:", model.coef_) # You should now see 4 coefficients
```

Task 3: Performance Comparison (R^2 and RMSE)

In machine learning, adding relevant features should generally improve the model's performance.

Your Task: Compare the results of the baseline model (2 features) with your expanded model (4 features).

1. **R² Score:** Does adding Insulin and BP bring the R² score closer to **1.0**?
 2. **RMSE:** Does the average prediction error decrease with more inputs?
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Task 4: Health Science Interpretation

In a clinical setting, we must understand the "weight" of each input.

Your Task: Look at the printed coefficients (b_1 , b_2 , b_3 , b_4).

- Which feature has the **highest** positive coefficient?
- **Interpretation:** Does this mean that feature is the strongest predictor of high glucose in your dataset? Write a 2-sentence summary of your findings.